Effectiveness of Home Based Cognitive Training Programme on Quality of Life and Caregiver Burden among Elderly Persons with Mild Cognitive Impairment in Selected Community Settings of Puducherry

R. Govindakumari[1], Vijayalakshmi[2], Sai Sailesh Kumar Goothy[3], Vijay Raghvan[4]

[1]Ph.D Scholar, Saveetha Institute of Medical and Technical Sciences, Chennai, Tamil Nadu, India, [2]Principal, Vignesh Nursing College, Thiruvannamalai, Tamil Nadu, India, [3]Associate Professor, Department of Physiology, R.D. Gardi Medical College, Ujjain, M.P, India, [4]Research Director, Saveetha Institute of Medical and Technical Sciences, Chennai, Tamil Nadu, India

Abstract

Ageing is a complex process that decreases the ability of different body systems. It effects both physical and psychological functions and decreases the quality of life. When elderly people were weak physically, this will increase the caregiver burden. Home based cognitive therapy was found to be effective in the management of cognitive functions in the elderly patients. The present study was conducted at selected community settings of Puducherry.A total of 314 elderly participants were recruited for the study after obtaining the written informed consent.After recruiting, the participants were randomly grouped into two groups that is control and intervention groups with 157 participants in each group. Home based cognitive training program was administered to the intervention group individuals. The results of the study supports that the home based cognitive training programme is beneficial in the management of elderly population with mild cognitive impairment. There was significant improvement in the quality of life of the elderly patients and also significant decrease in the care givers burden followed by the intervention. The study recommends further detailed studies in this area to adopt the home based cognitive training programme along with routine medical treatment.

Keywords: *Cognition, Home based cognitive training program, Quality of life, Caregivers.*

Introduction

Ageing is a complex process that decreases the ability of different body systems. It effects both physical and psychological functions and decreases the quality of life. When elderly people were weak physically, this will increase the caregiver burden.[1] Increase in the age increases the probability of Decline in the cognition functions deteriorates the general health and decreases the quality of life. Home based cognitive therapy was found to be effective in the management of cognitive functions in the elderly patients.[2] In the patients with cognitive impairment, the major cognitive functions affected are attention, spatial memory and verbal memory.[3] These cognitive domains play a very important role in day today life of an individual. Home based cognitive therapy is simple and effective therapy that helps to restore majority of the cognitive functions.[4] It is easy to apply and patient friendly as he does not need to stay in the hospital. It decreases the financial burden on the patient and his care givers. Hence, the stress experienced by the care givers of the elderly person will be relieved.[4] It also reduces other burdens like time

Corresponding Author:
R. Govindakumari
Ph.D. Scholar, Saveetha Institute of Medical and Technical Sciences, Chennai, Tamil Nadu, India

and effort of the care givers. Earlier studies reported multiple benefits followed by the intervention. Though drug therapies are available, there are not able to restore the cognitive functions effectively and also associated with side effects.[5] Hence, there is a need for alternative therapy like home based cognitive therapy. The present study was undertaken to observe the efficacy of the home based cognitive training programme on quality of life and caregiver burden among elderly persons with mild cognitive impairment in selected community settings of Puducherry.

Materials and Method

Study Design: Experimental study

Study Setting: The present study was conducted at selected community settings of Puducherry.

Study Participants: A total of 314 elderly participants were recruited for the study after obtaining the written informed consent.Convenient sampling technique was used in the selection of the participants. The following criteria were used while recruiting the participants.

Inclusion Criteria:

1. Client with Age ≥ 60
2. Client whose score in MMSE greater than 20
3. Clients who are a permanent resident of the locality for at least in the past 6 months.
4. Clients who are willing to participate in the study

Exclusion Criteria:

1. Clients with significant neurological deficit
2. Client who are in antidepressants
3. Clients with history of alcohol abuse
4. Clients with significant physical disability

After recruiting, the participants were randomly grouped into two groups that is control and intervention groups with 157 participants in each group.

Home Based cognitive Training Programme:

Each training session comprised the following steps

1. **Orientation in time and space:** Performed using external aids such as a calendar and the day's newspaper, whereby participants determined the current day, month and year.
2. **Presentation of the Names:** Participants and the researcher presented their names. Verbal associations were elicited between the person's name and their appearance.
3. **Visual and Auditory Attention Exercises:** The visual attention task entailed identifying details in photographs, letters or figures amongst several stimuli spread out on a single printed sheet, as well as spotting differences between two similar photographs, among other activities. The auditory attention exercises included tasks such as detecting words in songs and identifying whether two consecutively repeated sequences of numbers or words matched or differed.
4. **Memory Exercises Using Visual Aids:** The categorization strategy with ecological tasks was used, i.e. simulated activities of daily living, graded from simple to more complex. Grocery items were used in the early stages of training progressing to supermarket item lists only, by the end of training.
5. **Transfer Task:** Practical tasks from activities of daily living were used, such as to the supermarket, giving and checking change, etc. The older adults were expected to calculate the total cost of the purchase and offer change or check the change received.

Cognitive training was given to all eligible subjects individually two sessions weekly continuously for a period of 1 month i.e. a total of 8 sessions. Each session will take approximately 1 hour.

Ethical Considerations: The present study protocol was approved by the institutional human ethical committee. The study was conducted as per the guidelines of ICMR.

Data Analysis: Data was analysed using SPSS 20.0. Data was expressed in frequency and percentage and mean and SE. Paired and unpaired t test was used to observe the significance of difference. Probability value less than 0.05 was considered as significant.

Results

Results were presented in table no 1. There was a significant decrease in the care giver burden in the experimental group when compared with the control group. There was a significant improvement in the quality of life in the individuals of experimental group when compared with control group.

Table 1: Care giver burden and quality of life of control and experimental groups.

S.No.	Parameter	Group	Mean±SE	Paired 't'-test		Un paired 't'-test	
				Con- pre test -1st post test	Experimental pre -1st post	Con- pre test– experimental pre test	Control post – experimental post
1.	Care giver burden	Control pre test	55.36±0.45	t=2.22 p=0.03*		t=1.98 p=0.048*	t=60.11 p<0.001***
		Control 1st post test	53.87±0.49				
		Experimental pre test	54.14±0.43		t=65.55 p<0.001***		
		Experimental 1stpost test	19.61±0.29				
2.	Overall Quality of life	Control pre test	180.88±1.67	t=0.678 p=0.498		t=0.272 p=0.785	t=42.73 p<0.001***
		Control 1st post test	182.36±1.82				
		Experimental pre test	180.24±1.64		t=48.03 p<0.001***		
		Experimental 1stpost test	279.73±1.37				

*P<0.05 is significant; **P<0.01 is significant: ***P<0.001 is significant

Discussion

The present study was undertaken to observe the efficacy of the home based cognitive training programme on quality of life and caregiver burden among elderly persons with mild cognitive impairment in selected community settings of Puducherry. The results of the study supports that the home based cognitive training programme is beneficial in the management of elderly population with mild cognitive impairment. There was significant improvement in the quality of life of the elderly patients and also significant decrease in the care givers burden followed by the intervention. The study recommends further detailed studies in this area to adopt the home based cognitive training programme along with routine medical treatment. It was reported that about 15 million people are providing care giving to the elderly population.[6,7] Interestingly, majority of these care givers are family members who are sparing their time and losing their earnings and undergoing immense stress. Further, hospitalization is most common in the elderly patients with impairment of cognition.[8] During hospitalization, physical, psychological, social and financial aspects of burden is involved.[9] Impairment of cognition in elderly persons is most common and further if untreated, it can be converted as severe diseases like Alzheimer's disease.[10] The available drugs and other medical treatments are costly and difficult to afford them by the average individual. Hence, the need for a non-pharmacological therapies are essential for the management of the impairment of cognitive functions.[11] Home based cognitive training program was considered as essential and better therapy in the management of the cognitive decline as it does not have contra indications and affordable.[12] Earlier studies reported that home based cognitive training program improves the quality of life of the participants.[13] It also increases the memory and cognition of the patients so that their dependency on the care givers will be decreased. This will reduce the burden on the care givers.[14.15] Interestingly, home based cognitive training program was found to be very effective in improving the cognition in both the healthy and patients with cognitive decline.[16-18] The present study agrees with earlier studies as we have observed improvement in the quality of life and decline in the care givers burden.

Conclusion

The results of the study supports that the home based cognitive training programme is beneficial in the management of elderly population with mild cognitive impairment. There was significant improvement in the quality of life of the elderly patients and also significant decrease in the care givers burden followed by the intervention. The study recommends further detailed studies in this area to adopt the home based cognitive training programme along with routine medical treatment.

Conflicts of Interest: None declared.

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